NERON'S BLOWUP OF A SMOOTH-AT-P ALGEBRA IS SMOOTH AT P' WITH AN EXPLICIT COTANGENT SEQUENCE

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Editorial note. Input lemma-neron-functorial is same-chapter NOT carded (dropped in judgment as a pure pointer to Algebra blowup-base-change) and is declared external.

Allowed standard prerequisites (editorial).

- Functoriality of Neron's blowup. lemma-neron-functorial (same chapter, NOT carded, declared external). Permits replacing A by localizations off p throughout the proof.
- Base change of smooth ring maps. Algebra, lemma base-change-smooth (cross-chapter, named). A/pi A is smooth over kappa at p.
- Blowup along a regular sequence. More on Algebra, lemma blowup-regular-sequence (cross-chapter, named). Gives the presentation A' = A[y_1,...,y_c]/(pi y_i - g_i).
- Naive cotangent exact sequence for a surjection. Algebra, lemma exact-sequence-NL and lemma NL-surjection (cross-chapter, named). Computes H_1(NL_{A'/R}) and the conormal map of the blowup presentation.
- Independence of differentials of parameters. Algebra, lemma computation-differential (cross-chapter, named). Gives the kappa(p)-linear independence of dg_i from separability of the residue extension.

Source reference. Original reference 030O: focused-seeds/geometry/S9-smo-neron-blowup-smooth/ref-tags.txt, source lines 5–5. Navigation pointer only: the counted proof cites this statement by \ref as a standing input; the deduction the counted proof has to make is carried out inside the counted span, and no part of that deduction is delegated to this pointer.

Source reference. Original reference 00T2: focused-seeds/geometry/S9-smo-neron-blowup-smooth/ref-tags.txt, source lines 6–6. Navigation pointer only: the counted proof cites this statement by \ref as a standing input; the deduction the counted proof has to make is carried out inside the counted span, and no part of that deduction is delegated to this pointer.

Source reference. Original reference 07BP: focused-seeds/geometry/S9-smo-neron-blowup-smooth/ref-tags.txt, source lines 7–7. Navigation pointer only: the counted proof cites this statement by \ref as a standing input; the deduction the counted proof has to make is carried out inside the counted span, and no part of that deduction is delegated to this pointer.

Source reference. Original reference 00T4: focused-seeds/geometry/S9-smo-neron-blowup-smooth/ref-tags.txt, source lines 8–8. Navigation pointer only: the counted

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Source reference. Original reference 00TT: focused-seeds/geometry/S9-smo-neron-blowup-smooth/ref-tags.txt, source lines 9–9. Navigation pointer only: the counted proof cites this statement by \ref as a standing input; the deduction the counted proof has to make is carried out inside the counted span, and no part of that deduction is delegated to this pointer.

Source reference. Original reference 00TU: focused-seeds/geometry/S9-smo-neron-blowup-smooth/ref-tags.txt, source lines 10–10. Navigation pointer only: the counted proof cites this statement by \ref as a standing input; the deduction the counted proof has to make is carried out inside the counted span, and no part of that deduction is delegated to this pointer.

Source reference. Original reference 00RV: focused-seeds/geometry/S9-smo-neron-blowup-smooth/ref-tags.txt, source lines 11–11. Navigation pointer only: the counted proof cites this statement by \ref as a standing input; the deduction the counted proof has to make is carried out inside the counted span, and no part of that deduction is delegated to this pointer.

Source reference. Original reference 00S2: focused-seeds/geometry/S9-smo-neron-blowup-smooth/ref-tags.txt, source lines 12–12. Navigation pointer only: the counted proof cites this statement by \ref as a standing input; the deduction the counted proof has to make is carried out inside the counted span, and no part of that deduction is delegated to this pointer.

Source reference. Original reference 00NQ: focused-seeds/geometry/S9-smo-neron-blowup-smooth/ref-tags.txt, source lines 13–13. Navigation pointer only: the counted proof cites this statement by \ref as a standing input; the deduction the counted proof has to make is carried out inside the counted span, and no part of that deduction is delegated to this pointer.

Source reference. Original reference 061L: focused-seeds/geometry/S9-smo-neron-blowup-smooth/ref-tags.txt, source lines 14–14. Navigation pointer only: the counted proof cites this statement by \ref as a standing input; the deduction the counted proof has to make is carried out inside the counted span, and no part of that deduction is delegated to this pointer.

Source reference. Original reference 07BU: focused-seeds/geometry/S9-smo-neron-blowup-smooth/ref-tags.txt, source lines 15–15. Navigation pointer only: the counted proof cites this statement by \ref as a standing input; the deduction the counted proof has to make is carried out inside the counted span, and no part of that deduction is delegated to this pointer.

Source reference. Original lemma 0BJ3: sources/stacks-smoothing/pinned-original-source.tex, source lines 1124–1124. Navigation pointer only: the counted proof cites this statement by \ref as a standing input; the deduction the counted proof has to make is carried out inside the counted span, and no part of that deduction is delegated to this pointer.

Source reference. Original reference 0BIQ: focused-seeds/geometry/S9-smo-neron-blowup-smooth/ref-tags.txt, source lines 16–16. Navigation pointer only: the counted

proof cites this statement by \ref as a standing input; the deduction the counted proof has to make is carried out inside the counted span, and no part of that deduction is delegated to this pointer.

Source reference. Original reference 0BJ2: sources/stacks-smoothing/pinned-original-source.tex, source lines 1085–1085. Navigation pointer only: the counted proof cites this statement by \ref as a standing input; the deduction the counted proof has to make is carried out inside the counted span, and no part of that deduction is delegated to this pointer.

Lemma 0.1. *In Situation 0BJ2 assume that $R \to A$ is smooth at $\mathfrak{p}$ and that $R/\pi R \subset \Lambda/\pi\Lambda$ is a separable field extension. Then $R \to A'$ is smooth at $\mathfrak{p}'$ and there is a short exact sequence*

$$0 \to \Omega_{A/R} \otimes_A A'_{\mathfrak{p}'} \to \Omega_{A'/R,\mathfrak{p}'} \to (A'/\pi A')^{\oplus c}_{\mathfrak{p}'} \to 0$$

where $c = \dim((A/\pi A)_{\mathfrak{p}})$.

Proof. By Lemma 0BJ3 we may replace A by a localization at an element not in $\mathfrak{p}$; we will use this without further mention. Write $\kappa = R/\pi R$. Since smoothness is stable under base change (Algebra, Lemma 00T4) we see that $A/\pi A$ is smooth over κ at $\mathfrak{p}$. Hence $(A/\pi A)_{\mathfrak{p}}$ is a regular local ring (Algebra, Lemma 00TT). Choose $g_1, \ldots, g_c \in \mathfrak{p}$ which map to a regular system of parameters in $(A/\pi A)_{\mathfrak{p}}$. Then we see that $\mathfrak{p} = (\pi, g_1, \ldots, g_c)$ after possibly replacing A by a localization. Note that $\pi, g_1, \ldots, g_c$ is a regular sequence in $A_{\mathfrak{p}}$ (first π is a nonzerodivisor and then Algebra, Lemma 00NQ for the rest of the sequence). After replacing A by a localization we may assume that $\pi, g_1, \ldots, g_c$ is a regular sequence in A (Algebra, Lemma 061L). It follows that

$$A' = A[y_1, \ldots, y_c]/(\pi y_1 - g_1, \ldots, \pi y_c - g_c) = A[y_1, \ldots, y_c]/I$$

by More on Algebra, Lemma 0BIQ. In the following we will use the definition of smoothness using the naive cotangent complex (Algebra, Definition 00T2) and the criterion of Algebra, Lemma 07BU without further mention. The exact sequence of Algebra, Lemma 00S2 for $R \to A[y_1, \ldots, y_c] \to A'$ looks like this

$$0 \to H_1(NL_{A'/R}) \to I/I^2 \to \Omega_{A/R} \otimes_A A' \oplus \bigoplus\nolimits_{i=1,\ldots,c} A'\mathrm{d}y_i \to \Omega_{A'/R} \to 0$$

where the class of $\pi y_i - g_i$ in I/I^2 is mapped to $-\mathrm{d}g_i + \pi \mathrm{d}y_i$ in the next term. Here we have used Algebra, Lemma 07BP to compute $NL_{A'/A[y_1,\ldots,y_c]}$ and we have used that $R \to A[y_1, \ldots, y_c]$ is smooth, so $H_1(NL_{A[y_1,\ldots,y_c]/R}) = 0$ and $\Omega_{A[y_1,\ldots,y_c]/R}$ is a finite projective (a fortiori flat) $A[y_1, \ldots, y_c]$-module which is in fact the direct sum of $\Omega_{A/R} \otimes_A A[y_1, \ldots, y_c]$ and a free module with basis $\mathrm{d}y_i$. To finish the proof it suffices to show that $\mathrm{d}g_1, \ldots, \mathrm{d}g_c$ forms part of a basis for the finite free module $\Omega_{A/R,\mathfrak{p}}$. Namely, this will show $(I/I^2)_{\mathfrak{p}}$ is free on $\pi y_i - g_i$, the localization at $\mathfrak{p}$ of the middle map in the sequence is injective, so $H_1(NL_{A'/R})_{\mathfrak{p}} = 0$, and that the cokernel $\Omega_{A'/R,\mathfrak{p}}$ is finite free. To do this it suffices to show that the images of $\mathrm{d}g_i$ are $\kappa(\mathfrak{p})$-linearly independent in $\Omega_{A/R,\mathfrak{p}}/\pi = \Omega_{(A/\pi A)/\kappa,\mathfrak{p}}$ (equality by Algebra, Lemma 00RV). Since $\kappa \subset \kappa(\mathfrak{p}) \subset \Lambda/\pi\Lambda$ we see that $\kappa(\mathfrak{p})$ is separable over κ (Algebra, Definition 030O). The desired linear independence now follows from Algebra, Lemma 00TU. □